

Database Fundamentals

A database is an organized collection of structured data stored in a computer system. Databases are essential for storing, retrieving, and managing data efficiently and reliably.

Types of Databases:

1. Relational Databases: Store data in tables with rows and columns.

Examples: PostgreSQL, MySQL, Oracle, SQL Server

Key concept: ACID properties (Atomicity, Consistency, Isolation, Durability)

2. NoSQL Databases: Flexible schema for unstructured data.

Examples: MongoDB, Cassandra, Redis

3. Graph Databases: Store data as nodes and relationships.

Examples: Neo4j

PostgreSQL is an advanced open-source relational database system first released in 1996. It supports JSON, arrays, and full-text search.

SQL Basics:

SQL (Structured Query Language) is the standard language for database queries:

- SELECT: retrieve data
- INSERT: add new records
- UPDATE: modify existing records
- DELETE: remove records

Example query:

```
SELECT name, age FROM users WHERE age > 18 ORDER BY name;
```

Indexes improve query performance by creating pointers to data rows.

A database transaction is a sequence of operations treated as a single unit.

If any operation fails, all changes are rolled back to maintain consistency.